



Paper ID : 261733

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Subject Code:DPH106AT

Roll No:

PHARM D
(SEM. I) THEORY EXAMINATION 2025-26
REMEDIAL MATHEMATICS

TIME: 3 HRS

M.MARKS: 70

Note: Attempt all Sections. In case of any missing data; choose suitably.

SECTION A**1. Attempt all questions in brief.****02 x 7 = 14**

Q no.	Question
a.	Define identity matrix with example.
b.	The slope of the line joining the points (2,-5) and (4,1) is ?
c.	In a triangle, a=6, b=8, and included angle C=60°. Find side c
d.	Differentiate: $y = x^3 + 5x^2 - 2x + 7$
e.	Evaluate: $\int_0^1 x^2 dx$
f.	If $A = \begin{bmatrix} 1 & -1 \\ 2 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 1 \\ 4 & -1 \end{bmatrix}$, find A + B
g.	Write Laplace transform of t^n

SECTION B**2. Attempt any three of the following:****07 x 3 = 21**

a.	Solve by Cramer's rule : $2x + y = 5$, $3x - 2y = 4$
b.	In $\triangle ABC$, $A=45^\circ$, $B=60^\circ$, $a=8$, find b, c
c.	Write the co factor of the elements of the determinant $\begin{vmatrix} 3 & -1 & 1 \\ 1 & 4 & 7 \\ 4 & 2 & 3 \end{vmatrix}$
d.	Differentiate: $y = e^x \sin x$
e.	Find Laplace transform of $f(t)=e^{2t} \sin t$

SECTION C**3. Attempt any one part of the following:****07 x 1 = 07**

a.	If $A = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 1 \\ 4 & 2 \end{bmatrix}$, find AB and BA.
b.	Find the inverse of $A = \begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{bmatrix}$

4. Attempt any one part of the following:**07 x 1 = 07**

a.	Find equation of tangent to circle $x^2 + y^2 = 25$, at point (3,4).
b.	Evaluate: $\lim_{x \rightarrow 0} \frac{\sin x}{x}$



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5. Attempt any *one* part of the following:

07 x 1 = 07

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|----|--|
| a. | Verify Euler's theorem for $u = x^2 + y^2$ |
| b. | Using Leibnitz theorem, find nth derivative of $y = x^2 e^x$ |

6. Attempt any *one* part of the following:

07 x 1 = 07

- | | |
|----|---|
| a. | Evaluate using integration by parts: $\int x \sin x \, dx$ |
| b. | Solve variable separable equation $\frac{dy}{dx} = \frac{x^2}{y}$ |

7. Attempt any *one* part of the following:

07 x 1 = 07

- | | |
|----|--|
| a. | Solve: $D^2y - 4Dy + 4y = 0$. Where $D = \frac{dy}{dx}$ |
| b. | Find inverse Laplace transform of $\frac{1}{s^2+9}$ |